# Sanjay Kumar Vamula

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## SUMMARY


- **Data Engineer / Analyst** with **3+ years of experience** in building scalable **ETL pipelines**, **real-time data workflows**, and **BI solutions** across various domains including technology, industrial, and business intelligence.
- Proficient in **Python (Pandas, NumPy, scikit-learn)**, **SQL/PLSQL**, and **R**, applying **data cleaning**, **statistical modeling**, and **machine learning** techniques to solve complex business problems and drive data-driven insights.
- Strong expertise in **ETL orchestration** using tools like **Apache Airflow**, **PySpark**, **Azure Data Factory**, **SSIS**, and **Databricks**, automating **high-volume pipelines** that process **1M+ records monthly**.
- Skilled in developing **interactive dashboards** and **KPI-driven insights** using **Tableau, Power BI, Qlik**, and **Alteryx**, enabling decision-making for executives across **finance, operations, and compliance**.
- Hands-on experience with **cloud data platforms** including **AWS (Redshift, Glue, S3, Lambda, Athena)**, with exposure to **Azure (ADF, Synapse, ML)** and **GCP (BigQuery, Dataflow)** for scalable data processing and storage solutions.
- Delivered high-impact projects such as **modeling predictive analytics**, **energy consumption forecasting**, and **business operations analysis**, converting complex datasets into actionable recommendations that align with strategic goals.
- Knowledgeable in **DevOps practices** (Git, Docker, CI/CD), **Agile/Scrum methodologies**, and **regulatory compliance standards** (HIPAA, SOX), ensuring that data solutions are secure, reliable, and compliant.


## PROFESSIONAL EXPERIENCE

**BCBS** — **Wichita, KS**
**Jr. Data Engineer** — **Mar 2025 – Present**

- **Architected** end-to-end ETL workflows in **Python, SQL**, and **Apache Airflow** to ingest and transform **1.2M+** monthly healthcare claims, reducing data latency from 24 hours to under 3 hours and enabling near real-time analytics.
- **Designed** and optimized **complex SQL** queries and PostgreSQL views to audit claims submission accuracy, detecting and preventing over 8,000 erroneous records per quarter to maintain compliance with **HIPAA and internal** governance rules.
- **Developed** Python and **Pandas-based** automation for multi-step data validation, eliminating over 140 manual processing hours per month and freeing analysts to focus on higher-value investigations.
- **Partnered** with senior analysts and business leaders to blueprint and deliver **Power BI dashboards** that monitor claim cycle times, revenue performance, and customer touchpoints—accelerating leadership decision cycles by 48 hours.
- **Implemented** proactive data pipeline monitoring with AWS CloudWatch and **CI/CD-integrated** GitHub Actions, reducing incident response time from **2 hours to 15 minutes** and ensuring uninterrupted reporting for downstream systems.

**Dell Technologies** — **India**
**Data Engineer** — **Aug 2021 – Dec 2022**

- Architected high-volume, low-latency **streaming data pipelines** using **Apache Kafka** and **AWS Kinesis** to process over **2M daily events**, enabling **real-time analytics** and supporting latency-sensitive reporting across various business sectors.
- Designed and implemented end-to-end **ETL workflows** using **Apache Airflow**, **dbt**, and **Python**, reducing data processing time from **24 hours to under 3 hours** for **1.2M+ monthly transactional records**.
- Developed scalable transformation logic in **Python (Pandas, NumPy)** and **Spark SQL** to integrate and standardize datasets from **15+ internal and external sources**, ensuring analytical accuracy and consistency across all systems.
- Built and maintained **dimensional data models** in **Amazon Redshift** and **Snowflake**, providing a unified semantic layer that supported **finance**, **compliance**, and **executive decision-making dashboards**.
- Instituted automated **data quality** and **compliance checks** using **Great Expectations** and custom **Python validation frameworks**, preventing ingestion of **8,000+ invalid records per quarter** and ensuring data integrity.
- Implemented proactive pipeline observability using **Prometheus**, **Grafana**, and **AWS CloudWatch**, reducing incident recovery time from **90 minutes to under 20 minutes** through automated alerts and root cause analytics.
- Collaborated with cross-functional teams, including business leaders, data analysts, and machine learning engineers, to design data products that reduced **predictive model training time** and accelerated **KPI reporting** by **48 hours**.
- Mentored junior engineers on best practices for **workflow orchestration**, **schema design**, and **data governance**, improving team delivery capacity and reducing onboarding time for new hires.

**Adani** — **India**
**Data Analyst** — **Jun 2020 – Jul 2021**

- **Architected** SQL-based data models for logistics and supply chain datasets exceeding **10M records**, isolating high-impact anomalies that informed **multimillion-dollar** inventory and distribution decisions.
- **Engineered** interactive KPI dashboards in **Power BI and Tableau**, enabling executives to track shipment lead times, order fulfillment rates, and cost-per-route metrics in real time.
- **Designed** and executed **Python (Pandas, NumPy) pipelines** to cleanse and standardize raw operational data from **15+ disparate** sources, eliminating duplications and ensuring compatibility with downstream analytics.
- **Directed** requirement-gathering sessions with operations, finance, and IT teams to translate business objectives into technical specifications, driving adoption of a company-wide analytics platform.
- **Deployed** automated Python scripts and **Excel VBA solutions** generated **50+ recurring** reports monthly, reducing turnaround times from day to hours and freeing analyst capacity for strategic projects.

- **Produced** executive-ready operational efficiency reports that pinpointed **$1.2M annual** cost avoidance opportunities and informed optimization of vendor contract terms.
- **Structured** training-ready datasets for machine learning pipelines, enabling accurate demand forecasting models with a **12-month prediction** horizon.
- **Authored** detailed data workflow documentation mapping data ingestion, transformation logic, and governance controls, ensuring compliance with **SOX and internal** audit standards.

## TECHNICAL SKILLS

- **Programming & Analytics:** Python (Pandas, NumPy, Seaborn, Machine Learning Models)**,** SQL/PLSQL (Joins, CTEs**,** Window Functions**)**, R (Statistical / HEOR Analysis), Data Cleaning, A/B Testing, Hypothesis Testing, Regression, ARIMA Forecasting, Statistical Analysis
- **Data Analysis & Visualization:** Tableau (Dashboard Development), Power BI (DAX, Power Query), Qlik Sense, QlikView, MS Excel (Advanced Functions, PivotTables), SSRS, Alteryx
- **ETL, Data Engineering & Pipelines:** Databricks, Azure Data Factory, SSIS, Apache Airflow, PySpark, Hadoop, Spark, Automated Data Pipelines, API Integration, Data Modeling
- **Databases & Warehousing:** PostgreSQL, MySQL, Oracle, AWS Redshift, Azure SQL Database, Google BigQuery, MongoDB, Cassandra, Redis
- **Cloud Platforms:** AWS (S3, Redshift, Glue, EC2, Lambda, Athena)
- **DevOps & Infrastructure:** Git, Docker
- **Project Tools & Methodologies:** JIRA, Confluence, Agile (Scrum), SDLC, UAT
- **Exposure / Familiar With:** Azure (Machine Learning, Data Factory, Synapse), GCP (BigQuery, Dataflow), Kubernetes, Terraform, Jenkins, Flask, FastAPI, dbt, Kafka, Great Expectations, Prometheus, Grafana

## EDUCATION

**Wichita State University**, **Wichita, KS**
Master of Science in Data Science, **Jan 2023 – Dec 2024**

**Geethanjali College of Engineering & Technology**, **Keesara, TS**
Bachelor of Engineering in Electronics & Communication Engineering **Sep 2017 – May 2021**

## PROJECTS

### Predictive Analysis of FEMA Assistance Eligibility

- Processed and optimized **20.9M FEMA records** in **Databricks**, applying data casting, null handling, and partitioning into Parquet to improve SQL query performance and scalability.
- Automated ingestion and transformation workflows using **Apache Airflow** and **PySpark**, simulating production-grade pipelines.
- Built **logistic regression models** with **scikit-learn**, **Matplotlib**, and **Seaborn** to predict disaster relief eligibility; presented insights and policy recommendations at **KDSC 2024**.

### Texas Energy Dynamics Analysis

- Engineered a **20-step Tableau Prep workflow** with **Airflow pipelines** to clean and merge **26M hourly electricity pricing and generation records**, integrating weather and energy datasets from **AWS S3**.
- Applied **PySpark** and **SQL** for time-series and volatility analysis, uncovering demand spikes, renewable intermittency, and vulnerabilities linked to extreme weather.
- Designed **interactive Tableau dashboards** with **LOD/WINDOW calculations** and scenario filters, enabling regulators to simulate demand–supply shocks.

### Fan Engagement & Demographic Shift Analysis

- Analyzed attendee survey data using **Python (Pandas, NumPy, Matplotlib)** for cleaning, statistical modeling, and comparative analysis of demographic and behavioral trends.
- Leveraged **AWS Redshift** and **SQL** for scalable data storage and querying; built **Tableau dashboards** to visualize shifts in audience demographics and satisfaction drivers.
- Applied **correlation and multiple regression models** to identify enjoyment as the strongest predictor of satisfaction; delivered executive-style insights guiding targeted marketing and retention strategies.